



REVIEW

Why are clouds white, but rain clouds dark?

REFERENCES

13 verified

TOKENS

≈2.0k

VERIFICATION

Crossref

COMPILED

27 Aug 2026

SUMMARY

A cloud looks white because its tiny clear drops or bits of ice send the Sun's colors back to us mixed together. A thick rain cloud blocks and loses more light before it reaches the bottom, so the part we see can look grey or dark.

Picture a small white cloud on a sunny day, then a dark ceiling of cloud just before rain. They can contain the same clear water. The puzzle is not what color the water is. It is what happens to sunlight on its journey through the cloud.

01 What is a cloud actually made of?

An ordinary warm cloud is a crowd of tiny clear water drops. Aircraft measurements show that a usual drop is about **22 millionths of a metre across**. That is much wider than a wave of visible light.^{1,2}

Because a drop is this large, it sends red, green, and blue light away with almost the same strength, although the

exact directions can differ.^{3,4} High, cold clouds use pieces of ice instead. Ice also sends light in many directions, while its shape and roughness change the pattern.^{5,6}

02 How can clear drops make a white cloud?

One clear drop does not paint light white. But a cloud contains a huge crowd of drops. Sunlight turns again and again as it meets them, so its colors leave in many directions while still mixed together. To our eyes, that mixed light is a soft white glow.^{7,8,3} Figure 1 is a helper picture for that journey.

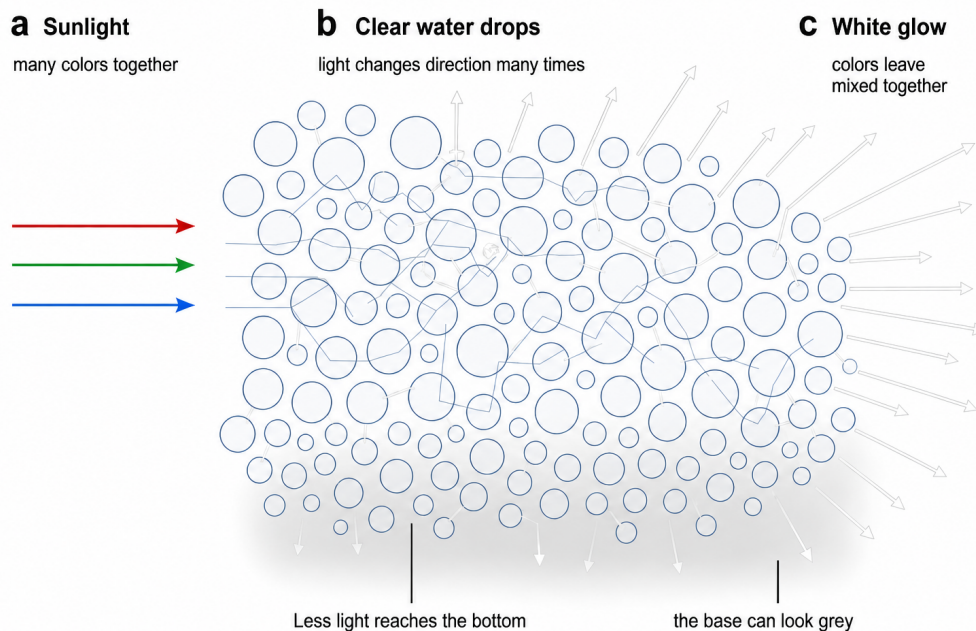


Figure 1. A crowd of clear drops makes a white cloud. Sunlight goes into the cloud, and the drops send it in many new directions. Much of the light escapes as a white glow. Less reaches the bottom of a thick cloud. The lines show the main idea, but real light does not follow these exact drawn paths.^{7,3,2,9}

03 Then why does a rain cloud look dark?

A cloud gets harder for light to cross when it holds more water, has more drops, or gives the light a longer path. In measurements from **13 overcast days**, the amount of liquid water explained most of the changes in how hard the cloud was for light to cross.^{2,10}

In a deep cloud, much of the light escapes from the top and sides before it can reach the bottom. The base looks grey because it is dimmer, not because the drops have turned grey or black.^{7,3} Real clouds are also lumpy, so the Sun's position and your viewing position decide which bright and shaded parts you see.^{9,1}



04 But are clouds always white or grey?

Here is the twist: white and grey are usual, but not the only answers. Measurements under thick cloud found that the daylight was not perfectly colourless. After many trips through water drops, a tiny bit more red light was lost, so the light below became a little bluer.¹¹

Thin clouds can also make coloured rings near the Sun when their drops are nearly the same size. Those colours wash into a bright grey background as the cloud gets thicker.¹² Some cloud edges show narrow rainbow-like bands. A recent study found that these bands can come from the drop size changing along the edge.¹³

05 What is the answer you can tell a friend?

A white cloud and a dark rain cloud can be made from the same clear water. Tell your friend: **the drops stay clear; the light's journey changes.** A thinner cloud sends plenty of mixed-colour light toward us, so it looks white. A thicker cloud lets less light reach its base, so that part looks grey or dark.^{7,3}

The helper picture can now step aside. For any real cloud, the remaining question is how its particular drop sizes, ice shapes, thickness, sunlight, and viewing angle combine to make the exact shade you see.

References

13 · VERIFIED VIA CROSSREF

1. Fu D, Di Girolamo L, Rauber RM, McFarquhar GM, Nesbitt SW, Laveridge J, Hong Y, van Diedenhoven B, Cairns B, Alexandrov MD, Lawson P, Woods S, Tanelli S, Schmidt S, Hostetler C, Scarino AJ (2022) An evaluation of the liquid cloud droplet effective radius derived from MODIS, airborne remote sensing, and in situ measurements from CAMP 2 Ex. *Atmospheric Chemistry and Physics*. <https://doi.org/10.5194/acp-22-8259-2022>
2. Kim B, Schwartz SE, Miller MA, Min Q (2003) Effective radius of cloud droplets by ground-based remote sensing: Relationship to aerosol. *Journal of Geophysical Research: Atmospheres*. <https://doi.org/10.1029/2003jd003721>
3. Kokhanovsky A (2004) Optical properties of terrestrial clouds. *Earth-Science Reviews*. <https://doi.org/10.1016/s0012-8252%2803%2900042-4>
4. Bréon F, Goloub P (1998) Cloud droplet effective radius from spaceborne polarization measurements. *Geophysical Research Letters*. <https://doi.org/10.1029/98gl01221>
5. Yang P, Bi L, Baum BA, Liou KN, Kattawar GW, Mishchenko MI, Cole B (2013) Spectrally Consistent Scattering, Absorption, and Polarization Properties of Atmospheric Ice Crystals at Wavelengths from 0.2 to 100 μm . *Journal of the Atmospheric Sciences*. <https://doi.org/10.1175/jas-d-12-039.1>
6. Yang P, Hioki S, Saito M, Kuo CP, Baum BA, Liou KN (2018) A Review of Ice Cloud Optical Property Models for Passive Satellite Remote Sensing. *Atmosphere*. <https://doi.org/10.3390/atmos9120499>
- Bohren CF (1987) Multiple scattering of light and some of its observable consequences. *American Journal of Physics*. <https://doi.org/10.1119/1.15109>
- Davis AB, Marshak A (2010) Solar radiation transport in the cloudy atmosphere: a 3D perspective on observations and climate impacts. *Reports on Progress in Physics*. <https://doi.org/10.1088/0034-4885/73/2/026801>
- Kato S, Marshak A (2009) Solar zenith and viewing geometry-dependent errors in satellite retrieved cloud optical thickness: Marine stratocumulus case. *Journal of Geophysical Research: Atmospheres*. <https://doi.org/10.1029/2008jd010579>
- Brenguier JL, Pawlowska H, Schüller L, Preusker R, Fischer J, Fouquart Y (2000) Radiative Properties of Boundary Layer Clouds: Droplet Effective Radius versus Number Concentration. *Journal of the Atmospheric Sciences*. <https://doi.org/10.1175/1520-0469%282000%29057%3C0803:rpoblc%3E2.0.co;2>
- Lee RL, Hernández-Andrés J (2005) Colors of the daytime overcast sky. *Applied Optics*. <https://doi.org/10.1364/ao.44.005712>
- Gedzelman SD, Lock JA (2003) Simulating coronas in color. *Applied Optics*. <https://doi.org/10.1364/ao.42.000497>
- Laven P, Hall L (2026) Iridescent colored bands on clouds. *Applied Optics*. <https://doi.org/10.1364/ao.580006> ■

00.